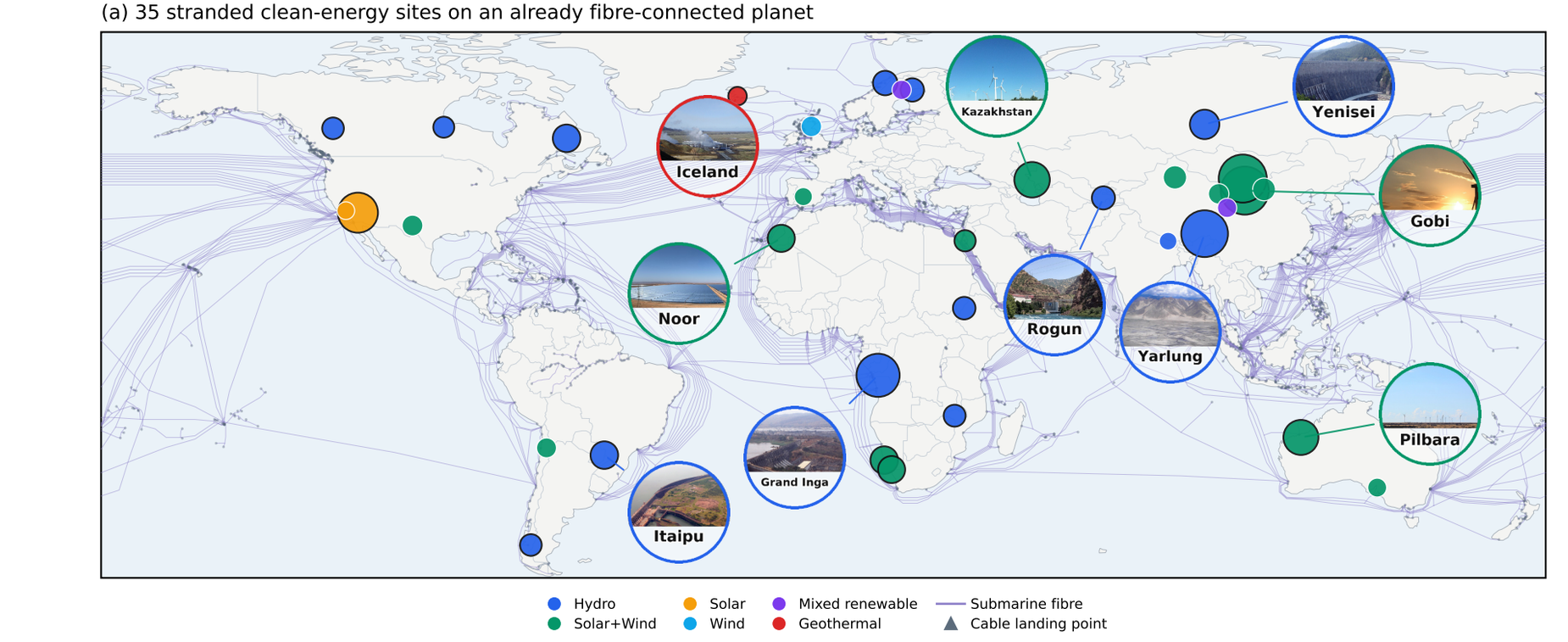
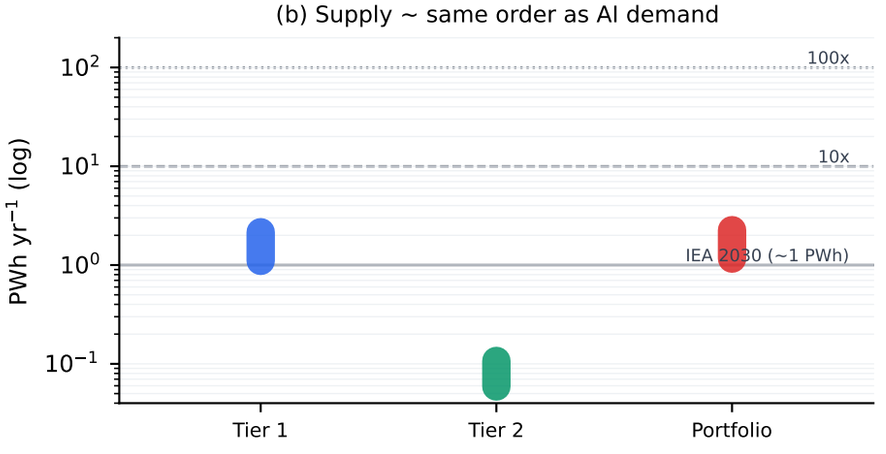
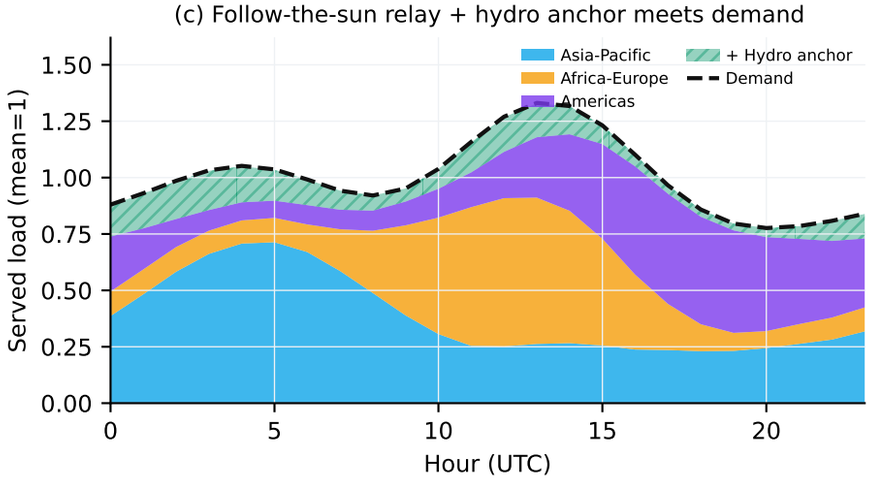
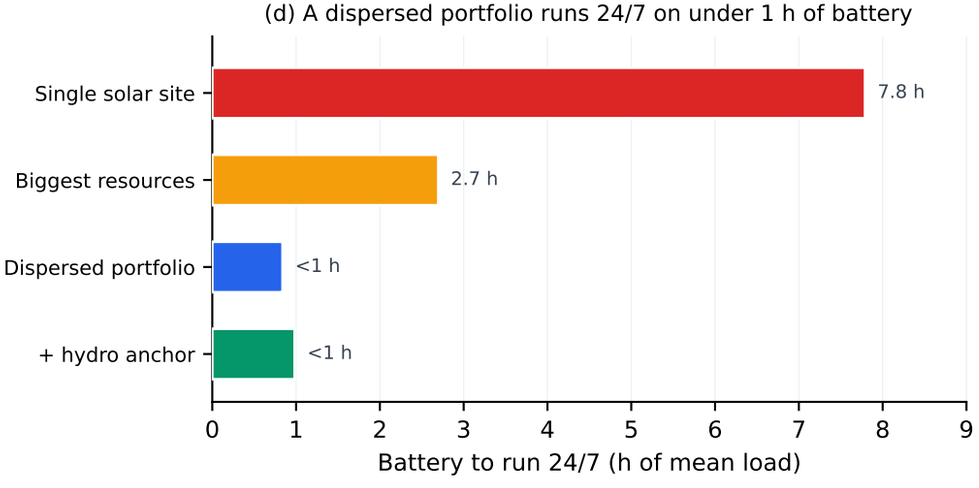
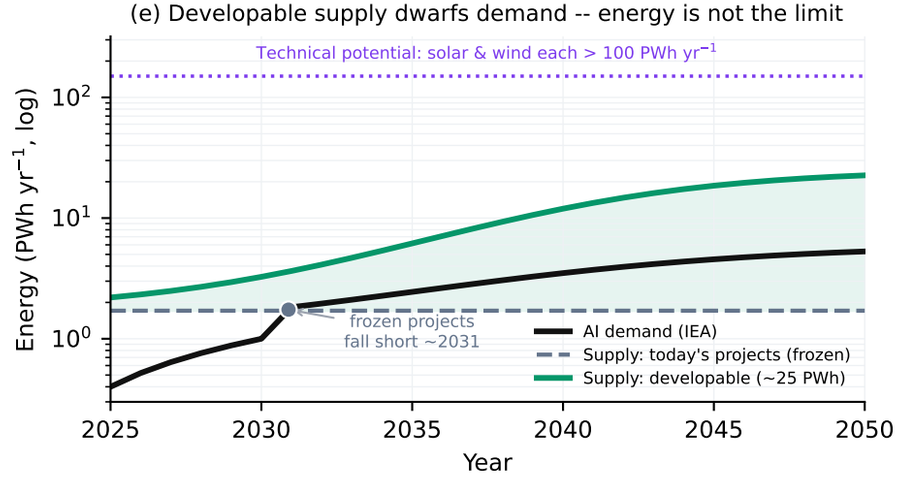

(a) 35 stranded clean-energy sites on an already fibre-connected planet
Iceland
Kazakhstan
Yenisei
Gobi
Noor
Rogun
Yarlung
Pilbara
Grand Inga
Itaipu
Hydro
Solar+Wind
Solar
Wind
Mixed renewable
Geothermal
Submarine fibre
Cable landing point


(b) Supply ~ same order as AI demand
PWh yr^{-1} (log)
100x
10x
IEA 2030 (~1 PWh)
Tier 1
Tier 2
Portfolio


(c) Follow-the-sun relay + hydro anchor meets demand
Asia-Pacific
Africa-Europe
Americas
+ Hydro anchor
Demand
Served load (mean=1)
Hour (UTC)


(d) A dispersed portfolio runs 24/7 on under 1 h of battery
Single solar site
Biggest resources
Dispersed portfolio
+ hydro anchor
7.8 h
2.7 h
<1 h
<1 h
Battery to run 24/7 (h of mean load)


(e) Developable supply dwarfs demand -- energy is not the limit
Technical potential: solar & wind each > 100 PWh yr^{-1}
Energy (PWh yr^{-1}, log)
frozen projects
fall short ~2031
AI demand (IEA)
Supply: today's projects (frozen)
Supply: developable (~25 PWh)
Year